

Rudransh Joshi

Portfolio: rudraxd.in
GitHub: github.com/FireHead90544

AI/ML Engineer
Email: work@rudraxd.in
LinkedIn: linkedin.com/in/rudraxd

EDUCATION

- Uttarakhand Technical University** Dehradun, India
• *Bachelor of Technology - Artificial Intelligence & Machine Learning — GPA: 8.74* Sep 2022 - Jul 2026
Courses: Artificial Intelligence, Machine Learning, Deep Learning, Data Science, Genetic Algorithms, Data Structures, Algorithms, Operating Systems, Databases, Networking, Computer Organization & Architecture, Automata Theory, Compiler Design, Digital Logic

SKILLS SUMMARY

- Languages:** Python, TypeScript, C, Go, SQL, Bash
- AI/ML:** PyTorch, TensorFlow, ONNX, LLMs, LangChain, NumPy, Pandas, Matplotlib, Scikit-Learn, Gymnasium
- Frameworks:** Django, FastAPI, Next.js, React, TailwindCSS, Qt, OpenCV, FFMpeg
- DevOps:** Git, GitHub, CI/CD, GNU/Linux, AWS/GCP/Azure, Cloudflare, Nginx, Docker
- Embedded:** Arduino, ESP32, Raspberry Pi, FreeRTOS

EXPERIENCE

- MiyuLabs** Hybrid — Noida, India
• *Founder / Software & Systems Engineer* Nov 2025 - Present
 - Architected and deployed MiyuLabs' end-to-end **IoT platform**, spanning ESP32/FreeRTOS firmware, Go + PostgreSQL + MQTT backend, Cloudflare R2, and Next.js web infrastructure, with **secure provisioning, OTA updates, offline sync, and CI/CD**; **averaging 150K+ web requests from ~2.4K visitors weekly**.
 - Engineered MiyuMail and Dispatch, implementing **self-hosted email infrastructure & durable campaign delivery** with IMAP sync, idempotent state, rate-limited batching, retries, crash recovery, and cross-platform native packaging.
- Confidential Client** Remote — Delhi, India
• *ML Engineer (Freelance)* Jan 2026 - Apr 2026
 - Built and deployed a computer vision system for automated quality assessment in a large-scale supply chain workflow, training a custom Ultralytics/YOLOv26 segmentation model on ~3K hand-labeled samples **achieving mAP50-95 score of ~0.93** with FP16 quantization to enable **real-time batch-level analysis**.
 - Implemented an edge-first inference pipeline using ONNX Runtime Web with API fallback and developed an analytics dashboard supporting **near-real-time detection under ~400ms** and data-driven QA decisions.
- E Giraffes Digital Pvt. Ltd.** Remote — Pune, India
• *AI Automation Engineer (Intern)* Jun 2024 - Aug 2024
 - Designed standardized AI asset-generation workflows, **reducing onboarding time by ~30% & manual intervention** across creative pipelines; automated SharePoint/Teams approvals via Power Automate, **cutting manual effort by 50%**.
- Cyber Efficient** Remote — Texas, USA
• *Full Stack Developer (Part-Time)* Dec 2020 - Feb 2022
 - Architected CyberEfficient.io with Django/Bootstrap, achieving **~35% faster load performance**, and developed a Discord bot **serving 10K+ users under ~20ms avg latency** with a Flask URL shortener handling **25K+ links**.

PROJECTS

- EFNet – Edge Face Network (PyTorch, CNN-ViT, ArcFace):** Researched and designed a lightweight **CNN-ViT hybrid** for few-shot open-set face recognition on edge devices, combining depthwise-separable convolutions, SE blocks, linear attention, and full MHSA across resolutions. Built a **~562K-parameter, ~86.5M-FLOP** backbone pretrained on **~494K CASIA-WebFace** images with ArcFace, enabling **fine-tuning-free ID recognition from as little as 1 ref. image**.
- RLSwarm – Physics-Based Swarm RL Environment (Python, Gymnasium, PyGame):** Researched and developed a continuous swarm foraging simulator built on Gymnasium. Benchmarked PPO and DQN agents in scalable multi-agent settings, demonstrating **zero-shot generalization from N → 2N agents** under fixed-area constraints, **retaining ~30% of baseline reward** under doubled agent density.
- while(MOO) – Offline Cattle Breed Identification System (PyTorch, Next.js):** Implemented an offline-first breed classification system using a **custom explainable CNN (MooNet)** with **~96.7% accuracy** built over MobileNetV4, with feature engineering & preprocessing optimized for edge inference via ONNX Web Runtime. Integrated a PWA dashboard with domain-specific chatbot (AskMoo) and structured breed knowledge base (MOOpedia) using FastAPI and Gemini API.
- how-cli – Context-Aware AI Terminal Assistant (Python, LangChain, Typer):** Developed a cross-platform CLI assistant converting **natural-language prompts into OS-specific sequential shell commands**, supporting local/cloud LLMs with shell, package manager, and project context detection with added **safety guardrails** for destructive commands.

PUBLICATIONS

- Joshi et al., Policy Optimization for Scalable Swarm Robotics in Physics-Based Environments:** Under review, 2026. Multi-Agent Systems, Reinforcement Learning, Swarm Intelligence.

HONORS AND AWARDS

- 1st Place, Smart India Hackathon - 2023, 2025
- 2nd Place, UTU' Socio Hackathon - Apr 2025
- Best Overall Hack, Hackathon With Lewis - Oct, 2024
- Most Liked Hack, Codédex Summer Hackathon - Jul 2024
- Academic Excellence Award, School Topper (Secondary: 97%; Senior Secondary: 93%) - 2020, 2022